

The Advance Directive Ontology (ADO): A Computable Representation of End-of-Life Care Preferences in Advanced Heart Failure

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Abstract

End-of-life care preferences are recorded in instruments (advance health care directives, POLST forms, and code-status orders) that are low-resolution or disease-blind and support no automated reasoning over conditional, graded, negatable preferences. We present the Advance Directive Ontology (ADO), an OWL 2 ontology (67 classes, 22 properties) scoped to five advanced heart failure decision points (CPR, ventilation, ICD/device management, vasopressor/inotrope escalation, dialysis) with 21 intervention subclasses. A reasoner matches a patient’s encoded preferences against a clinical scenario and returns one of four outputs (clear, partial, no coverage, vague) while preserving original language. The reasoner scored 16/16 on a single-patient vignette suite and 35/36 (97%) on code-status/POLST mapping; a condition-blind ablation dropped to 12/16 (75%) and a pure-LLM baseline to 10–12/16, isolating the contribution of the structured layer. An LLM pipeline extracts preferences at F1 0.97 with zero hallucinations on out-of-scope text. Additionally, a multi-patient cohort stress test (50 profiles across seven encoding-quality levels, 25 scenarios, 3,550 decision cells) showed 41.4% oracle agreement overall versus 17.4% for flat checkboxes; the primary failure mode was false activation (34%) rather than direction error (1.3%). Because 83% of reference answers were inherently uncertain, a system that declines to guess is calibrated to the problem rather than failing it. Following expert review, we frame ADO as a decision-support aid built off of an advanced directive for advance-care-planning documentation, not a legal directive.

Introduction

Patients record end-of-life (EOL) care preferences through several distinct instruments, each occupying a different layer of the care pathway: advance directives (AD) capture durable values and wishes; Physician Orders for Life-Sustaining Treatment (POLST) translate them into portable medical orders; and clinician-entered code-status orders (full-code, do-not-resuscitate, do-not-intubate, do-not-escalate) are what is actually acted on at the bedside during an acute crisis. An estimated 67% of intensive-care patients lack decision-making capacity at EOL,¹ yet fewer than 37% of US adults have any directive on file,² and the documents that do exist are rarely interoperable with electronic health record (EHR) systems.³

A common framing holds that the core problem is that directives are unstructured free text. This is partly true: the best ADs, such as the California statutory form and the University of California directive, are largely structured checkbox instruments. However, the more fundamental problem is that ADs are low-resolution and disease-blind. AD checkbox forms collapse conditional, graded preferences (e.g. “I would accept a ventilator temporarily if there is a reversible cause, but never indefinitely”) into only a few options, and they often omit the device- and disease-specific decisions that dominate advanced heart failure (HF): implantable cardioverter-defibrillator (ICD) deactivation, left-ventricular-assist-device (LVAD) management, and inotrope escalation (approximately 200,000 HF patients in the US have ICDs,⁴ yet most generic AD templates do not mention them).

For this project, we investigate heart failure as heart failure is a revealing scope for this problem. This is because its trajectory is unpredictable (oscillating between stable and acute states so “terminal” is hard to define), it is device-dependent in a way that a generic template does not anticipate, and patients cycle repeatedly through the emergency department and ICU, generating frequent, time-pressured preference-interpretation events.⁵ To our knowledge, no published OWL ontology supports description-logic reasoning over conditional, negatable, graded EOL preferences: the HL7 PACIO Advance Directive Interoperability guide³ transports directive *documents* as FHIR resources but does not reason over them; POLST encodes *orders* as binary checkboxes that lose conditional statements; and the Stanford Letter Project⁶ preserves narrative voice but is not computable. As such, ADO bridges both the information and the inference layer in heart failure scenarios.

Objectives

Our project had the following goals:

1. Build a formal, computable ontology of EOL care preferences for advanced HF.

2. Build a pipeline that encodes patient preferences within the structure of the ontology
3. Implement a reasoner that matches encoded preferences to a clinical scenario and returns an output, and ensure that the reasoner emits an uncertainty decision rather than forcing an answer in complex scenarios.
4. Demo an end-to-end version of the pipeline that encodes ontology preferences directly from free-text ADs using large language models (LLMs).
5. Perform evaluations and ablation studies to measure performance
6. Gather feedback on the system from an expert clinical ethics reviewer

While we design ADO to be a structured and computable ontology of preferences derived from the AD, we emphasize that ADO is a decision-support and documentation aid rather than a legal advance directive.

Methods

Advance directive concept inventory

We first sought to ground our ontology in real-world document diversity, so we compiled and cataloged 50 advance directive templates across 8 categories (Table 1), drawing on state statutory forms, national repositories (CaringInfo,¹² National POLST¹³), and specialized populations. Analyzing these documents confirmed our hypothesis that HF-specific decisions (ICD deactivation, inotrope/vasopressor escalation) are almost entirely absent from standard AD templates.

Table 1. Advance directive concept inventory by category.

Category	Count	Examples
State-specific ADs (US)	10	California AHCD, Texas Directive, NY Health Care Proxy
POLST/MOLST forms	6	National POLST, CA POLST, NY MOLST
Specialized populations	4	Five Wishes, Voicing My Choices, Dementia-specific
Psychiatric/mental health	7	Bazelon Center PAD, SAMHSA guide
Religious/faith-based	9	Catholic (NCBC), Halachic, Islamic (IMANA)
Organizational	4	VA Form 10-0137, AAFP, Mayo Clinic
DNR/DNAR forms	5	Texas OOH-DNR, CA Prehospital DNR
International	5	UK ADRT (NHS), BC “My Voice”
Total	50	

Ontology design

The ADO contains 67 classes, 13 object properties, and 9 data properties, authored in Protégé⁸ using OWL2⁷ for its formal description-logic semantics and reasoner compatibility. It is scoped to the five HF decision points where directives are most often consulted (CPR/resuscitation, mechanical ventilation, ICD/device management, vasopressor/inotrope escalation, and dialysis), with 21 specific intervention subclasses (Figure 1). Each preference is a `PreferenceStatement` linking an `Intervention` to an `ActivationCondition` (NYHA functional class,¹⁰ clinical condition, reversible-cause flag, care context, and time bound), a `PreferenceStrength` (Absolute / Strong / Conditional / Weak), an `isNegated` boolean, and the patient’s verbatim language via `originalText`. Clinical concepts are grounded in SNOMED CT¹¹ for interoperability.

In order to make ADO amenable to complex scenarios, we incorporated vagueness as a first-class category. The `VaguePreference` subclass preserves irreducibly imprecise patient language (“no heroic measures”) via `originalText` while assigning it to the nearest formal class, so the system acknowledges what it cannot fully represent rather than forcing false precision. Additionally, exclusionary preferences are encoded with the `isNegated` flag and, where needed, `complementOf` class expressions, a deliberate accommodation of OWL’s open-world semantics.⁷

Preference-input and reasoning pipeline

We first built a pipeline to transform a JSON structured preference input into the language of the ontology and then run the reasoner over it for a clinical scenario. The pipeline, written in Python, instantiates OWL individuals from

ADO Intervention class hierarchy — parsed from the OWL ontology
five HF decision points and their intervention subclasses

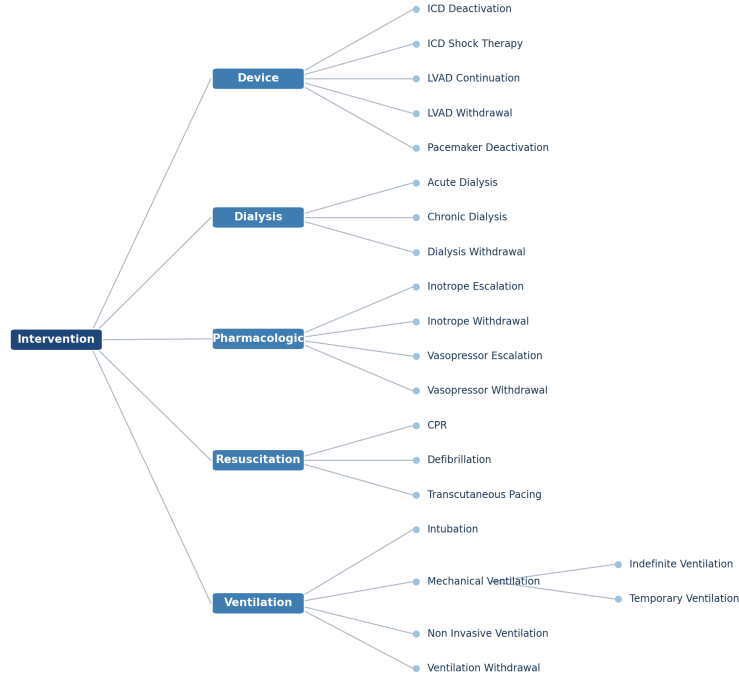


Figure 1. The Intervention class hierarchy, parsed directly from the OWL ontology: five HF decision points (blue) and their intervention subclasses, including the clinically meaningful temporary-vs-indefinite ventilation distinction.

structured JSON, validating inputs against controlled vocabularies for all 21 interventions, clinical conditions, NYHA classes, care contexts, and document types. It supports interactive CLI entry, JSON-file batch input, and example generation. A scenario-based query engine then takes the populated patient ontology plus a clinical scenario, matches each preference’s activation conditions against the scenario state, and classifies the result into one of four outputs (Table 2), which were chosen to force the system to be clinically honest, flagging uncertainty rather than forcing an answer.

Crucially, ADO supports conditional preferences, which are enforced by the reasoner. For example, a patient may express a temporal condition (e.g., “withdraw vasopressors if no improvement after 72 h”) or a care-context condition (e.g. treatment differences in hospice vs the ICU)—both of which are explicitly modeled into the ontology and obeyed by the reasoner.

Table 2. The reasoner’s four-category output.

Output	Meaning
Clear match	Activation conditions met; preference unambiguous
Partial match	Some conditions met, others unspecified or explicitly unmet; defer to surrogate
No coverage	The directive does not address this scenario
Vague match	Preference uses irreducibly imprecise terms; original language surfaced with nearest formal class

LLM extraction from free-text ADs

To show that ADO can ingest real free-text directives, we designed a system that can parse ADs in order to generate structured outputs, which can then be fed into the step described above. Specifically, we use Claude Opus 4.7 (Anthropic) with a Pydantic-typed schema to convert a verbatim directive paragraphs into the same JSONs the structured pipeline already accepts. Additionally, we use a closed-world approach in which the model is fed only what the patient explicitly stated and is instructed not to extrapolate or utilize outside knowledge.

Code-status and POLST derivation

Motivated by expert review (discussed below), which noted that the bedside is governed by code-status orders and POLST rather than the directive, we additionally built a layer that aggregates intervention-level preferences into a hospital code status. This allows the ADO to span the entire process of parsing an AD, converting it to a structured OWL format, emitting interventions proposed by a reasoner in response to a clinical situation, and generating code status orders and POLST. The system supports code status (Full code / DNR / DNI / DNR+DNI, with an optional “Do Not Escalate” qualifier), a POLST Section A order (Attempt CPR / DNAR), and a POLST Section B treatment level (Full / Selective / Comfort-Focused). Code status and Section A are scenario-grounded, so a conditional refusal yields DNR only when its conditions are met, and the system flags that conditionality, which a flat checkbox cannot express. Section B reflects standing willingness read from the preference set.

Evaluation design

We evaluated the system across six tracks:

- **Track 1 (vignettes):** 16 clinical vignettes spanning all five decision points and all four preference types (clear, conditional, negated, vague), with 10 additional held-out vignettes evaluated blind, scored against hand-authored gold labels.
- **Track 2 (extraction):** LLM extraction over 12 real directive template statements, including two deliberately out-of-scope statements (artificial nutrition/hydration; antibiotics) to test closed-world discipline.
- **Track 3 (code-status/POLST mapping):** 12 preference profiles derived from real inventory templates, scored against a gold standard adjudicated from POLST’s published definitional semantics and deliberately independent of ADO’s own logic.
- **Track 4 (structuring):** six messy narrative passages mixing vague wishes, hedged conditions, internal contradictions, surrogate delegation, and out-of-scope content, testing whether the pipeline produces auditable structure without fabricating missing information.
- **Track 5 (multi-patient cohort):** 50 patient profiles spanning seven encoding-quality levels (clean, typical, minimal, incomplete encoding, contradictory, outdated, and culturally complex), evaluated against 25 clinical scenarios covering canonical cases, boundary conditions, multi-condition overlaps, and scenarios intentionally outside directive scope, yielding 3,550 scored decision cells. Reference decisions came from a simplified oracle that applies preference-matching rules without ontology access, keeping the gold standard independent of the system under test; flat-checkbox and condition-blind baselines were scored in parallel.
- **Track 6 (architecture ablation):** We tested two ablation studies. The first was a condition-blind variant, which tested whether the explicit condition encoding in the ontology was useful. The second was to use an advanced LLM to predict interventions directly, bypassing the ontology entirely, in order to test the value of the ontology itself.

The ADO system was also reviewed by Dr. David Magnus, Director of the Stanford Center for Biomedical Ethics.

Results

Scenario-based reasoning on vignettes (track 1)

The reasoner achieved 16/16 (100%) decision accuracy and 16/16 (100%) match-type accuracy across the vignette suite for track 1 (Table 3), correctly handling all four output categories, temporal thresholds in both the “fires” and “does-not-fire-yet” directions (V10/V13), care-context gating (V5/V6), etiology-specific conditions (V14), and intervention granularity that prevents over-inference (V16). Two earlier failure modes documented in our progress report, a missing care-context property and a string-typed time bound, were resolved by the the care context and temporal conditioning we described earlier.

Extraction accuracy and closed-world discipline (track 2)

Over the 12 real-template statements (15 in-scope gold preferences), the extractor achieved precision 0.94, recall 1.00, F1 0.97, with negation, preference type (clear/conditional/vague), and conditionality each correct on 15/15 matched preferences. Importantly, it committed zero over-extractions on the two out-of-scope statements: given text about

Table 3. Clinical vignette evaluation. Match types: clear (9), partial (4), no coverage (2), vague (1).

ID	Clinical scenario	Intervention	Expected	System	Result
V1	Cardiac arrest, NYHA IV, no reversible cause	CPR	No (clear)	No (clear)	PASS
V2	Cardiac arrest, NYHA III, reversible cause	CPR	Partial	Partial	PASS
V3	Resp. failure, reversible cause	Temp. ventilation	Yes (clear)	Yes (clear)	PASS
V4	Ventilator 10d, no improvement	Indef. ventilation	No (clear)	No (clear)	PASS
V5	Enrolled in hospice	ICD deactivation	Yes (clear)	Yes (clear)	PASS
V6	In ICU, not hospice	ICD deactivation	Partial	Partial	PASS
V7	Cardiorenal syndrome, recovery expected	Acute dialysis	Yes (clear)	Yes (clear)	PASS
V8	Kidney failure, no recovery	Chronic dialysis	No (clear)	No (clear)	PASS
V9	Cardiogenic shock	Inotrope escalation	Yes (clear)	Yes (clear)	PASS
V10	Cardiogenic shock, vasopressors 96 h	Vasopressor w/d	Yes (clear)	Yes (clear)	PASS
V11	Respiratory failure	NIV (BiPAP)	Vague	Vague	PASS
V12	Stable, NYHA II	Pacemaker deactivation	No coverage	No coverage	PASS
V13	Cardiogenic shock, vasopressors 48 h	Vasopressor w/d	Partial	Partial	PASS
V14	Sepsis-AKI (not cardiorenal), recovery expected	Acute dialysis	Partial	Partial	PASS
V15	Stable, outpatient dialysis consult	Chronic dialysis	No (clear)	No (clear)	PASS
V16	VT, ICD about to shock	ICD shock therapy	No coverage	No coverage	PASS

artificial nutrition/hydration and antibiotics, both interventions outside our vocabulary, it extracted nothing rather than inventing a mapping into the nearest covered class. The single false positive (the only departure from perfect precision) was a defensible over-split: the vague “no heroic measures or to be kept alive by machines if I am dying” was mapped to two vague interventions (CPR and mechanical ventilation) where the gold expected one.

Code-status / POLST mapping (track 3)

Across the 12 directive profiles, ADO’s derived code status and POLST orders agreed with the POLST-semantics gold standard on 35/36 fields (97%), with 11/12 profiles matching on all three fields (Table 4). The hospital code-status confusion matrix was fully diagonal across all six observed categories (Full code, DNR, DNI, DNR/DNI, Do Not Escalate, DNR/DNI+Do Not Escalate) (Figure 2); chance-corrected agreement with the gold standard was Cohen’s $\kappa = 1.00$ for code status and POLST Section A, and $\kappa = 0.88$ for Section B.

The test also reveals that the condition-dependent ADO structure is obeyed through code status and POLST. We tested this by encoding the same directive for P9 and P10 (“no CPR if NYHA IV with no reversible cause”) and testing various scenario edge cases. Further, the single divergence (P5) was when ADO returned “Not specified” instead of presuming “full treatment”.

Table 4. Code-status / POLST mapping fidelity across 12 directive profiles (gold standard from POLST semantics).

Metric	Agreement
Hospital code status	12/12 (100%)
POLST Section A (resuscitation)	12/12 (100%)
POLST Section B (treatment level)	11/12 (92%)
Exact-profile match (all three fields)	11/12 (92%)
Overall field agreement	35/36 (97%)

Stably structuring messy, real-world directives (track 4)

We next ran six narrative passages through the extraction pipeline that contained vague wishes (“let me go peacefully”), hedged conditions, internal contradictions (“I’m a fighter . . . but I don’t want to be kept going just to exist”), surrogate delegation, and substantial out-of-scope content. From these, the layer produced 17 structured preferences (5 clear, 8 conditional, 4 vague), recovering 14/16 (88%) of the in-scope interventions while capturing the embedded conditions (reversible cause, hospice context, time bounds).

Critically, the passages included 11 out-of-scope topics, (faith, feeding tubes, organ donation, pet care, agent designation) yet resulted in no spurious orders (Figure 3). The model’s own notes surfaced the surrogate delegation in all three passages that contained it and flagged the genuinely ambivalent passage as irreducibly vague.

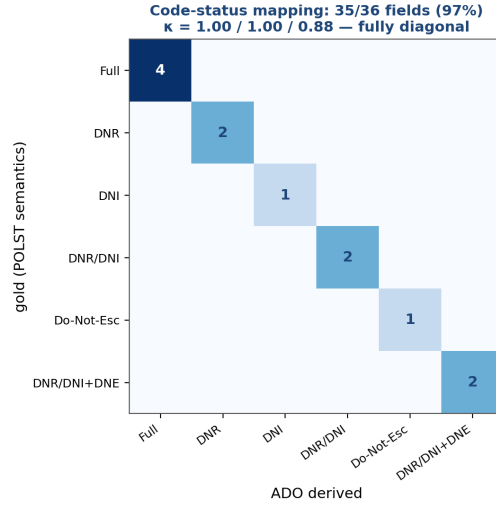


Figure 2. Hospital code-status confusion matrix (gold = POLST semantics): fully diagonal across all six observed categories.

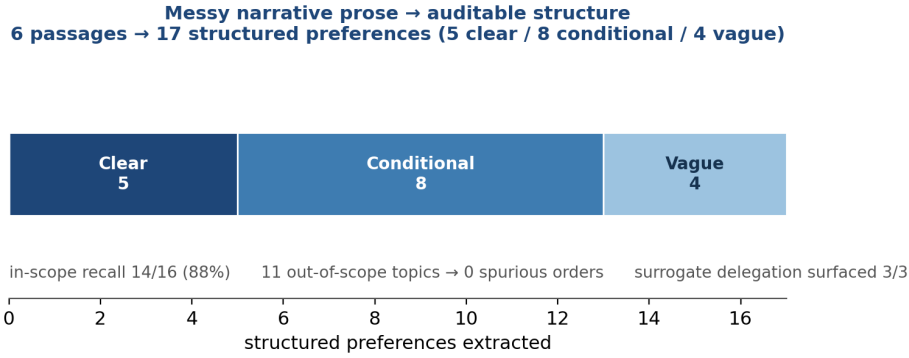


Figure 3. From messy narrative prose to auditable structure: six realistic passages yield 17 typed preferences, with out-of-scope content cleanly excluded and surrogate delegation surfaced.

Multi-patient cohort stress test (track 5)

We next ran a multi-patient cohort simulation with 50 independently constructed patient profiles, each with a distinct directive encoding, and evaluated against 25 clinical scenarios spanning canonical cases, boundary conditions (NYHA class thresholds, exact time-bound values), multi-condition overlaps, and scenarios explicitly outside the directive’s scope (pacemaker deactivation, LVAD malfunction, comfort-floor admission). This produced 3,550 scored decision cells.

ADO agreed with the reference oracle on 1,471/3,550 cells (41.4%), versus 38.3% for a condition-blind variant and 17.4% for a flat-checkbox baseline that treats any directive presence as a firm yes/no. Of the 3,550 reference answers, 83% were inherently uncertain (partial, no coverage, or vague); only 11% were clear “yes” and 6.5% clear “no.”

Performance stratified across all seven encoding levels as expected (Figure 4): clean profiles scored 48.3%, dropping to 36.2% for minimal, 38.7% for contradictory, and 29.9% for culturally complex patients. Two patterns are clinically informative. First, outdated directives (44.4%) outperformed contradictory ones (38.7%): temporal information, even when stale, gives the reasoner a tractable basis for inference, while internally conflicting preferences neutralize its conditional logic. Second, culturally complex patients showed zero ADO advantage over the condition-blind baseline (both 29.9%), because cultural and family-deference preferences require representational dimensions not yet in the ontology, a gap flagged by Dr. Magnus’s dimensionality critique. The flat checkbox collapsed most severely on incomplete-encoding and culturally complex patients (6.7% and 8.8%), confirming that structured encoding adds value

even when imperfect.

Scenario-type difficulty. Mean accuracy varied by scenario category: canonical cases 61%, multi-condition overlaps 53%, boundary conditions 43%, stress tests 38%, and out-of-scope scenarios 24%. The low out-of-scope figure is both expected, since the ontology was never designed to handle pacemaker deactivation or LVAD malfunction in isolation, and correct: returning no-coverage outside the directive’s scope is preferable to fabricating an answer.

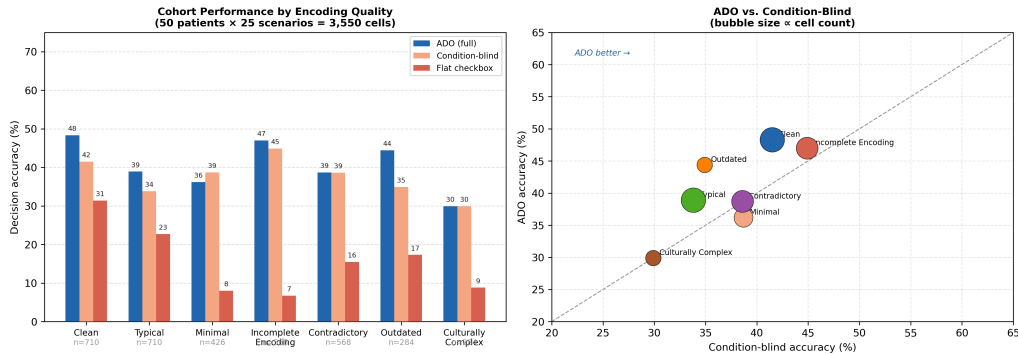


Figure 4. Cohort stress-test performance stratified by encoding quality (50 patients, 25 scenarios, 3,550 cells). Left: ADO, condition-blind, and flat-checkbox accuracy by encoding level; n per level shown below. Right: scatter of ADO vs. condition-blind per level; points above the diagonal show encoding-quality levels where the conditional logic adds value.

Ablation studies (track 6)

To isolate what the conditional layer contributes, we re-ran the suite with a condition-blind reasoner that fires any matching preference while ignoring activation conditions but keeps the ontology structure intact. Its decision accuracy fell to 12/16 (75%), failing the four conditional cases (V2, V6, V13, V14) (Figure 5). These failures are false-confident: the reasoner returns a definitive “yes/no” where the directive only conditionally applies, for example recommending ICD deactivation for an ICU patient whose preference was hospice-only. The activation-condition logic is what converts that dangerous over-reading into an honest “partial / defer to surrogate.” The 10 held-out vignettes, not used during system development, also achieved 10/10 (100%) decision accuracy, confirming that the full vignette performance is not an artifact of development-set overfitting.

We additionally tested whether the ontology and reasoner add value beyond asking an LLM to read a directive and answer clinical questions directly, bypassing extraction and reasoning altogether. We investigated two settings: (1) a minimal setting giving the model only the verbatim directive text and the clinical question, approximating a clinician pasting the directive into a chatbot; and (2) a full-context setting whose prompt is explicitly aligned to the reasoner’s four-category semantics and decision rules. Decision accuracy was 10/16 (62%) in the minimal setting and 12/16 (75%) in the full-context setting (Figure 5). The failure pattern is the diagnostic result: the minimal setting failed on every vignette with a ground-truth “partial” label, since it cannot represent conditional, non-binary outcomes; the full-context prompt recovered most of those but regressed on “vague” cases, forcing structure where the patient’s language was irreducibly imprecise. Neither LLM-only arm reached the full system’s 16/16. The ontology and reasoner therefore add value precisely on the complex, non-binary cases (conditional and vague preferences) that a free-text LLM either collapses into a confident yes/no or cannot honestly flag.

Failure mode analysis

To understand how/when ADO fails, we classified each of the 2,079 failure cells into five categories: (1) false activation, where ADO returns a definitive decision (yes/no/partial) but the oracle says the directive does not address the scenario (no coverage); (2) coverage gap, where ADO returns no coverage but the oracle expects a decision; (3) boundary confusion, where direction is correct but the confidence level (clear vs. partial) disagrees; (4) direction error, where ADO says yes when the oracle says no, or vice versa; and (5) correct.

Figure 6 shows that most errors come from false activations: 1,194 cases, or 33.6% of all cells. The other failure types are less common: 452 coverage gaps (12.7%), 387 boundary confusions (10.9%), and 46 direction errors (1.3%). In other words, ADO is much more likely to read a directive as applying when it should not than to miss a preference that does apply. False activations occur 2.6 times as often as coverage gaps.

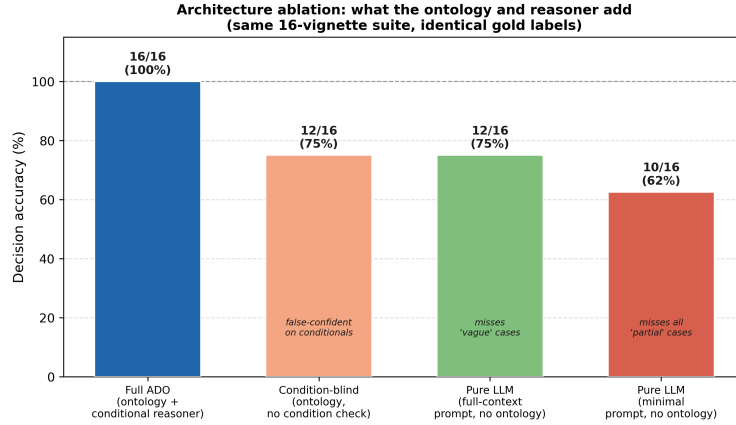


Figure 5. Architecture ablation on the 16-vignette suite. The full ADO pipeline (ontology + conditional reasoner) scores 16/16. Removing condition-checking but keeping the ontology drops to 12/16, with false-confident answers on conditionals. Removing the ontology entirely (a pure-LLM flow) scores 12/16 with an explicit reasoner-aligned prompt and only 10/16 with a minimal prompt, failing on the non-binary (partial, vague) cases the structured layer is designed to handle.

These errors are concentrated in the later scenarios, S20–S25, which include the out-of-scope and stress-test cases. Because this pattern appears across all encoding levels, the issue seems less about how the directive was encoded and more about the structure of the preferences themselves. When activation conditions are written broadly, the reasoner can apply them to clinical situations that the directive likely never meant to cover. Reassuringly, direction errors—the cases where ADO points in the wrong clinical direction—are rare, making up only 1.3

Taken together, the results suggest that ADO’s main risk is not that it reverses a patient’s stated wishes, but that it can sound too certain when the directive is actually silent. The most important fix is therefore to scope preferences more tightly: activation conditions should be more specific, and scenarios outside the five in-scope decision points should include explicit no-inference guards.

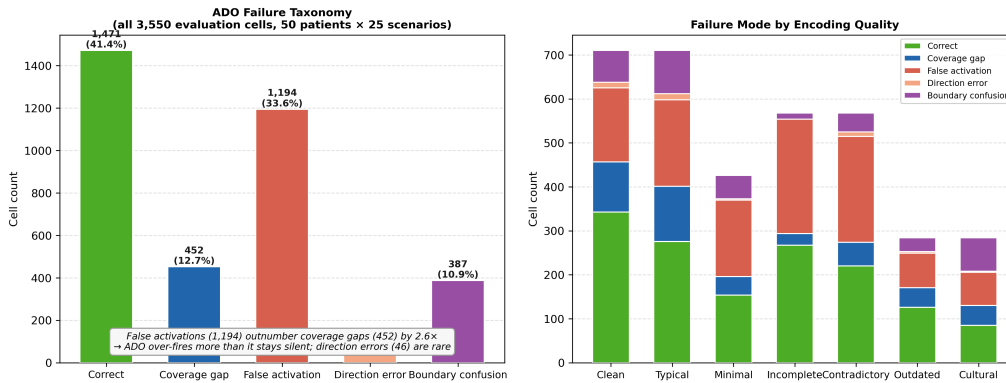


Figure 6. Failure mode taxonomy across all 3,550 cells. Left: absolute counts and percentages by failure category. Right: failure mode composition stratified by encoding-quality level; false activations (red) dominate across all levels, while direction errors (orange) remain rare throughout.

Decision-point coverage

For the five in-scope decision points the ontology provides 21 intervention classes, more granular than any single template in the inventory (standard ADs offer a binary on “mechanical ventilation,” whereas ADO distinguishes six ventilation-related interventions). Table 5 summarizes coverage and the gaps surfaced by expert review.

Table 5. Ontology coverage of common decision points.

Decision point	Coverage	Notes
CPR / resuscitation	Full (3 subclasses)	CPR, defibrillation, transcutaneous pacing
Mechanical ventilation	Full (6 subclasses)	Temporary vs. indefinite, invasive vs. non-invasive, withdrawal
ICD / device mgmt	Full (5 subclasses)	ICD deactivation, shock therapy, LVAD, pacemaker
Vasopressor / inotrope	Full (4 subclasses)	Escalation and withdrawal for both
Dialysis	Full (3 subclasses)	Acute, chronic, withdrawal
Artificial hydration/nutrition	Not yet covered	Flagged by expert review as a priority gap; addressed by POLST
Antibiotics, comfort care	Not covered	Future work

Expert review

Dr. Magnus provided valuable insight as to how the ADO system might fit into a large academic medical center such as Stanford Health Care. His central point was that the ADO “can’t [be thought of] as AHCDs from a legal perspective. The patient has not signed anything and it likely does not meet the regulatory requirements. This is more like a progress note.” He reaffirmed that “many AHCD directives . . . have more or less designed them as check boxes . . . it isn’t largely a problem of free text,” which sharpened our motivation toward the resolution-and-coverage argument above. His deepest critique was dimensional: “one problem with this system is that it is one dimensional. But patient preferences are often two or more dimensional. E.g., I don’t want to be intubated, but I want my family to be comfortable . . . so do what they say.” Under the law a capacitous patient’s preferences govern, but he noted that studies show most patients prefer that families be allowed to override. He flagged two concrete omissions, artificial hydration/nutrition and the POLST concept of a time-limited trial.

Following this, the ADO is well-suited to support the support an advance-care-planning (ACP) note. An ACP note in Epic documents a patient’s preferences regarding future medical care, living wills, and health care proxies; it is typically completed following a provider conversation with a patient about their understanding of their condition, their values, their surrogates, and their specific wishes. In this framing, structured information from an ADO is valuable as pre-populated fields in the ACP note, which can streamline completion and help ensure patient wishes are recorded even if a given topic is not explicitly discussed during the encounter. This naturally respects the prioritization of documented wishes: although preferences recorded in an advance directive may later be clarified or superseded through an ACP conversation, the ADO can preserve the advance directive as the best available representation of the patient’s wishes for any decision not revisited during the encounter.

Discussion

The evaluation supports our central claim: a description-logic ontology can represent conditional, graded, negatable EOL preferences and reason over them to produce a clinically honest output, including the honest non-answers (no coverage, vague) that a binary checkbox or a confident LLM cannot. Therefore, we advocate for use of the ontology with LLMs at the boundaries: the LLM extracts, the ontology is the auditable closed-world source of truth, and the reasoner produces the defensible four-category output. In high-stakes EOL care, a pure LLM is non-deterministic and prone to false confidence, whereas an auditable reasoner can honestly report when a directive is silent.

The cohort stress test also reinforces how the ADO can structure messy or vague inputs. Even though 41.4% overall oracle agreement appears low, 83% of reference answers are inherently uncertain, so any system that commits to a binary yes/no is wrong by construction on most cells, as the flat checkbox’s 17.4% confirms. ADO preserves uncertainty, which calibrates it to the actual distribution of real-world directive coverage.

The failure analysis shows that the dominant failure is false activation (34% of cells), rather than direction inversion (1.3%). While it is promising that the patients directions are not contradicted, this indicates the need to tighten the logic around out-of-scope scenarios. Nonetheless, if we remove these out-of-scope scenarios, accuracy on in-scope canonical scenarios increases to 61%.

In addition to the fact that ADO is not a legal AD and is best suited as a bridge between the AD and the ACP note, a number of other important considerations arose following speaking with Dr. Magnus. One such point is dimensionality—the current test coverage uses simple examples but real preferences can carry a second qualifier specifying who decides. The cohort’s culturally complex patients, which show zero ADO advantage over baseline, make this gap empirically

visible: family-deference and cultural framing cannot be represented in activation conditions alone. Further, Dr. Magnus pointed out that artificial hydration/nutrition should be added as a sixth decision point, the POLST time-limited trial should be modeled, and outputs should align with the real code-status taxonomy via an explicit no-escalation status.

Future work

Future work largely consists of continually validating and improving the ADO architecture and exploring integrations with existing EHRs and health systems. Following Dr. Magnus' recommendations as well as the advice of other experts, we can expand the decision points as well as classes in order to represent more possible clinical statements or reason over new clinical scenarios. Additionally, future work could look at real-world examples of the ACP note workflow by Stanford clinicians in order to understand how a prepopulated system might work. Finally, future work could expand the ontology, decision points, and the reasoner to other conditions outside of HF.

Limitations

One limitation of the evaluation is that the cohort oracle is a simplified rule-based simulation, not hand-adjudicated clinical gold. Further, OWL lacks native temporal reasoning, addressed here with numeric time bounds. Negation under the open-world assumption has edge cases we continue to document. Encoding fidelity (how completely the pipeline captures intent) is distinct from reasoning fidelity (how correctly the reasoner evaluates what was encoded), and the cohort shows that encoding gaps, not reasoning errors, drive most performance loss. Finally, the gold standards for extraction and mapping evaluations were team-authored; inter-rater agreement and validation against real-EHR code-status documentation (e.g., MIMIC-IV) remain future work.

Conclusion

We built a computable ontology of EOL care preferences for advanced heart failure that reasons over conditional, graded, negatable preferences and returns an intervention. The single-patient vignette suite showed 16/16 (100%) accuracy; a condition-blind ablation dropped to 12/16 (75%) and a pure-LLM baseline to 10–12/16, isolating what the structured layer contributes; and a multi-patient cohort stress test (50 profiles, 25 scenarios, 3,550 cells) showed 41.4% oracle agreement against a baseline of 83% inherent oracle uncertainty and 17.4% for flat checkboxes. A failure analysis showed that the primary failure is false activation (34%), the over-assertion of coverage rather than direction inversion (1.3%), which is more correctable and preferable to patient contradictions. Expert clinical-ethics review reframed ADO as a decision-support aid for advance-care-planning documentation and charted a path forward centered on multidimensional (who-decides) preference modeling, artificial nutrition/hydration coverage, and alignment with the bedside code-status system.

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Appendix

A. Code, files, and how to run

All code is available at https://github.com/patrickwalsh26/bmds210-AD_Ontology. The repository contains:

- `advanced_directives.owl`: base OWL ontology (67 classes, 22 properties).
- `ado/preference_input.py`: structured JSON → populated patient ontology pipeline.
- `ado/query_evaluation.py`: scenario-based reasoner; 16 development + 10 held-out vignettes.
- `ado/code_status.py`: aggregates preferences into hospital code status + POLST orders.
- `evaluation/track3_evaluation.py`: code-status mapping evaluation over 12 directive profiles.
- `evaluation/extraction_evaluation.py`: LLM extraction precision/recall over real directive text.
- `ado/llm_extraction.py`: closed-world free-text directive → structured JSON (Claude).
- `ado/llm_direct_reasoning.py`: pure-LLM direct-reasoning baseline (Gemini), no ontology.
- `evaluation/ablation_evaluation.py`: architecture ablation: full pipeline vs. LLM-only arms.
- `evaluation/realistic_simulation.py`: multi-patient cohort stress test (50 patients, 25 scenarios).
- `evaluation/failure_analysis.py`: failure mode taxonomy; five diagnostic figures.
- `evaluation/comprehensive_eval_suite.py`: full evaluation pipeline; generates all figures.
- `evaluation/data/patient_cohort.py`: 50 patient profiles across 7 encoding-quality levels.
- `evaluation/data/scenario_battery.py`: 25 clinical scenarios including out-of-scope stress tests.
- `evaluation/build_demo_owl.py`: builds 3-patient demo ontology for Protégé walkthrough.
- `evaluation/demo_query.py`: 30-second terminal demo showing conditional vs. blind reasoning.
- `data/populated/`: example populated ontologies and input JSON.

Requirements: Python 3.8+, a Java runtime (for owlready2’s Hermit reasoner), `pip install owlready2`; for the LLM pipeline, `pip install anthropic pydantic` and an `ANTHROPIC_API_KEY`.

Run the reasoner evaluation:

```
python -m ado.preference_input --example      # generate the example patient ontology
python evaluation/vignette_eval.py --suite all # 16 dev + 10 held-out vignettes
python evaluation/ablation_evaluation.py       # condition-blind + pure-LLM ablations
python evaluation/track3_evaluation.py         # code-status / POLST mapping
python evaluation/extraction_evaluation.py      # LLM extraction P/R/F1 (needs API key)
python evaluation/realistic_simulation.py       # 50-patient cohort stress test
python evaluation/failure_analysis.py          # failure-mode taxonomy + figures
```

B. Screenshots of program execution

Figure/Listing 1 shows the reasoner’s summary output; the per-vignette detail prints above it during execution.

Listing 1. `python evaluation/vignette_eval.py -suite all`: full vignette suite output.

```
=====
VIGNETTE EVALUATION
=====

Development (n=16)
Decision accuracy   : 16/16 (100%)
Match-type accuracy : 16/16 (100%)

Held-out (n=10)
Decision accuracy   : 10/10 (100%)
Match-type accuracy : 10/10 (100%)

No failures.
```

Listing 2. `python evaluation/realistic_simulation.py`: cohort stress test summary.

```
=====
MULTI-PATIENT COHORT SIMULATION (realistic / messy profiles)
=====
Cells evaluated      : 3550 (50 patients x 25 scenarios x interventions)

ADO decision vs ref : 41.4%
Condition-blind      : 38.3%
Checkbox (flat POLST, silent->full code) : 17.4%
Checkbox (silent->no coverage)             : 82.8%
```

ADO on cells WITH directive coverage : 27.9% (1228/3550 cells)
 Checkbox overconfident (nuanced gold -> flat yes/no) : 2929 cells

By messy_level:

clean	n=710	ADO 48.3%	blind 41.5%	checkbox 31.4%
typical	n=710	ADO 38.9%	blind 33.8%	checkbox 22.7%
minimal	n=426	ADO 36.2%	blind 38.7%	checkbox 8.0%
incomplete_encoding	n=568	ADO 47.0%	blind 44.9%	checkbox 6.7%
contradictory	n=568	ADO 38.7%	blind 38.6%	checkbox 15.5%
outdated	n=284	ADO 44.4%	blind 34.9%	checkbox 17.3%
culturally_complex	n=284	ADO 29.9%	blind 29.9%	checkbox 8.8%

Listing 3. python evaluation/failure_analysis.py: failure mode taxonomy.

```
=====
ADO Failure Mode Analysis
=====
Generated 3,550 evaluation cells

Failure taxonomy:
correct          : 1471 (41.4%)
coverage_gap     : 452 (12.7%)
false_activation : 1194 (33.6%)
direction_error  : 46 (1.3%)
boundary_confusion : 387 (10.9%)
```

Listing 4. python track3_evaluation.py: code-status mapping summary.

```
SUMMARY
Profiles          : 12
Exact-profile agreement : 11/12 (92%)
Code status       : 12/12 (100%)
POLST A          : 12/12 (100%)
POLST B          : 11/12 (92%)
Overall field agreement : 35/36 (97%)
(code-status confusion matrix fully diagonal; sole divergence P5 discussed in text)
```

Listing 5. python extraction_evaluation.py: LLM extraction summary.

```
Precision : 0.94   Recall : 1.00   F1 : 0.97
Per-field on 15 matched preferences: negation 15/15, type 15/15, conditionality 15/15
Over-extracted/hallucinated (out-of-scope FP) : 0  <- closed-world discipline holds
```

C. Screenshots of work completed

The ontology was authored in Protégé and queried programmatically with owlready2. Figure 7 shows the Intervention class hierarchy; Figure 8 shows a populated PreferenceStatement individual with its object- and data-property assertions.

D. Team member contributions

Member	Project contributions	Report contributions
Patrick Walsh	Ontology design and implementation in Protégé; concept inventory; structured preference-input pipeline; SNOMED CT grounding; LLM extraction pipeline; expert-review coordination	Primary author; wrote all sections; compiled references
Darren Chan	Repository setup and maintenance; scenario-based query engine; clinical vignette design; testing and gap analysis	Reviewed and edited; contributed evaluation results and analysis

E. Statement on sharing

We consent to the teaching team sharing this project as an example in future offerings of the course, with identifying information removed.

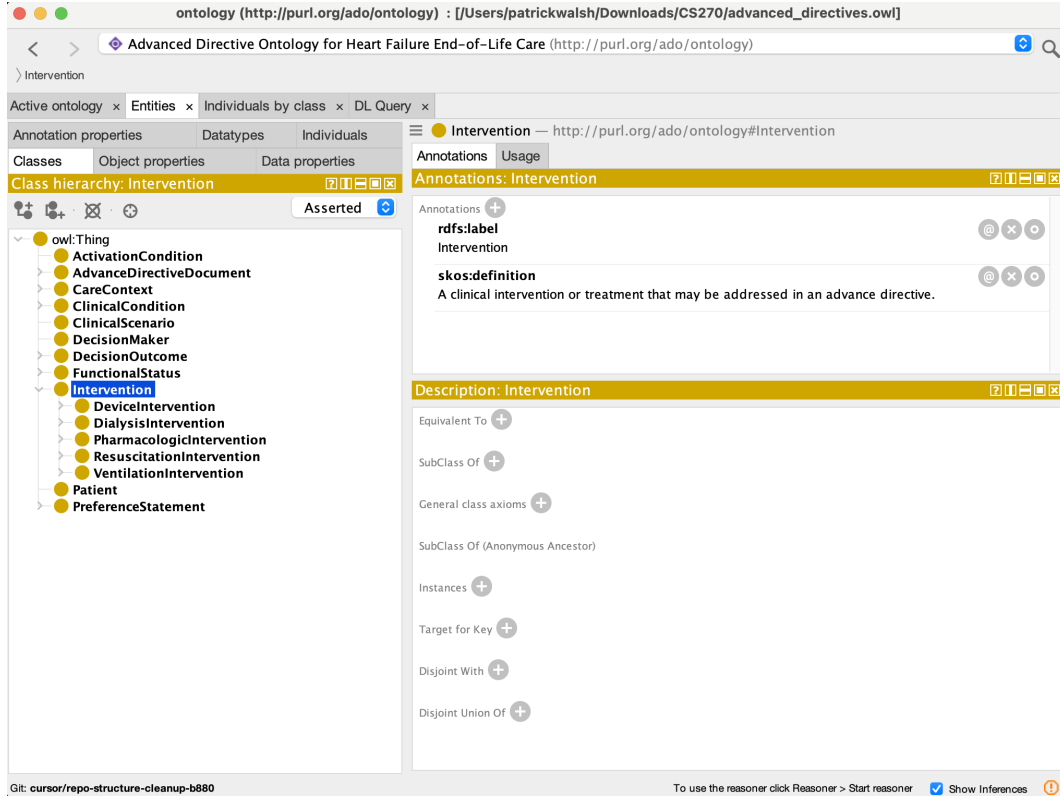


Figure 7. Protégé class hierarchy for the ADO ontology: the *Intervention* branch expands into five decision points (device, dialysis, pharmacologic, resuscitation, ventilation) and 21 intervention subclasses, including the clinically meaningful temporary-vs-indefinite ventilation distinction.

F. AI use statement

LLMs were used for two purposes.

1. As a system component. LLMs are a deliberate architectural element of ADO: a closed-world extraction pipeline (Claude Opus 4.7) converts free-text directives into structured JSON, and a pure-LLM direct-reasoning baseline (Gemini 3.5 Flash) was implemented as an ablation to quantify the value the ontology and reasoner add. In all system uses the ontology, not the LLM, is the auditable source of truth; LLM outputs are constrained by typed schemas and, in the extraction pipeline, by a closed-world prompt that forbids inference beyond the patient's stated text.
2. As an authoring aid. We used an AI coding and writing assistant (Claude) to help implement and debug evaluation scripts, generate scripts to produce publication-quality figures from computed results, and edit and reorganize prose in this report. No results were fabricated or generated by an LLM in place of running the evaluations.

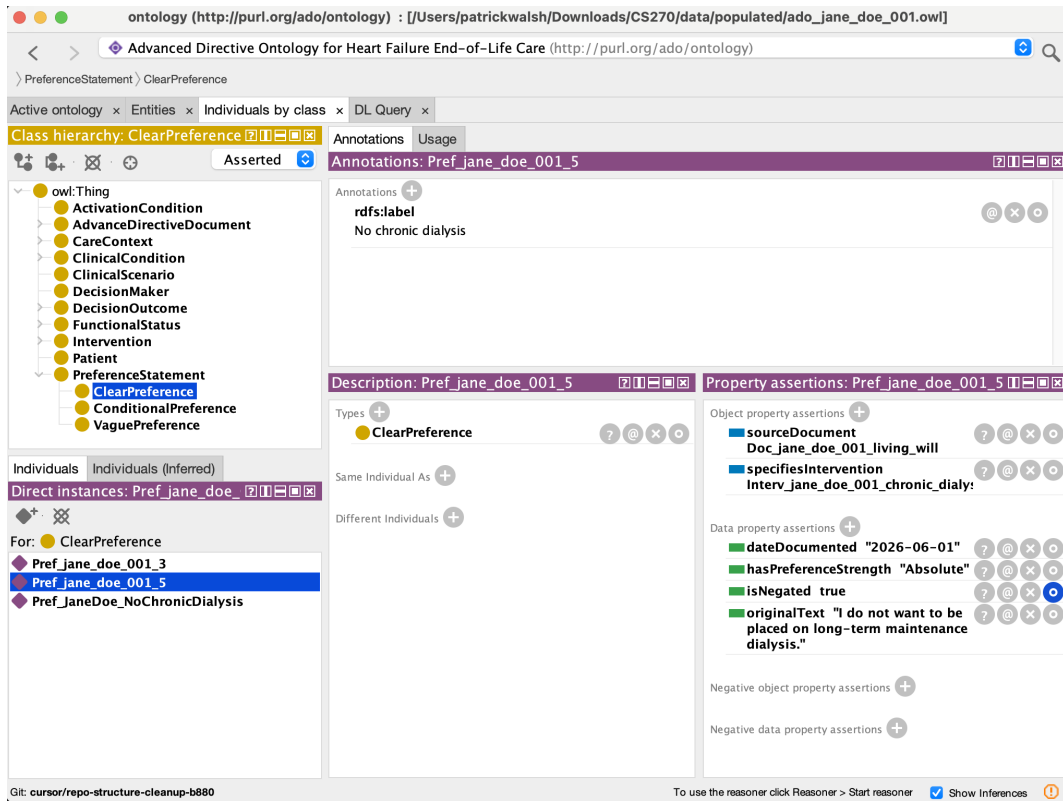


Figure 8. A populated `ClearPreference` individual (`Pref_jane_doe_001_5`, “No chronic dialysis”): object-property assertions link it to its source document and intervention, while data properties record negation (`isNegated = true`), strength (`Absolute`), documentation date, and the patient’s verbatim text.